Inverse Kaluza-Klein Scale Symmetry

Document I

An Alternative Solution Framework for the Schrödinger Equation via Scale-Projected Identity Decomposition

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Abstract

The standard Schrödinger equation operates on a wavefunction $\Psi(\mathbf{r}, t)$ defined over classical 3D space plus time. This document proposes an alternative framework in which the wavefunction is reinterpreted as the *scale-projected shadow* of a fuller identity structure $\mathcal{I}$ defined over an extended scale ladder $\{S_0, S_1, \ldots, S_D\}$. The standard solution is recovered at the S_3–S_4 projection; corrections at higher scales produce measurable residuals that map onto dark-sector and quantum-gravity phenomenology. The framework does not modify the Schrödinger equation's algebra; it reinterprets what the wavefunction *is* and why it succeeds and fails where it does.

Contents

§1 The Standard Framework and Its Projection Assumption

1.1 The Schrödinger Equation as Usually Stated

The time-dependent Schrödinger equation is:

$$i\hbar \frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \hat{H}\,\Psi(\mathbf{r},t) \tag{1}$$

where $\hat{H}$ is the Hamiltonian operator, $\mathbf{r} \in \mathbb{R}^3$, and Ψ is complex-valued. The Born rule assigns probability density:

$$\rho(\mathbf{r},t) = |\Psi(\mathbf{r},t)|^2 \tag{2}$$

This framework is extraordinarily predictive at the scale S_3–S_4 (spatial/binding + time-ordered). It fails or requires patching at:

- Very high energy / short distance: ultraviolet divergences
- Many-body regimes: exponential Hilbert space scaling
- Gravitational coupling: no consistent quantum gravity
- Cosmological scale: dark matter and dark energy residuals
- Measurement: wavefunction collapse has no agreed dynamics

Remark 1.1. These failure modes are not random. Under the present framework they are systematic projection losses: physics that exists at scales beyond S_4 but is not captured by the $\Psi(\mathbf{r},t)$ representation.

1.2 The Implicit Projection Assumption

The standard equation contains a hidden assumption:

$$\boxed{\Psi(\mathbf{r},t) \equiv P_{S_3 \cup S_4}(\mathcal{I})} \tag{3}$$

That is, the wavefunction *is* the identity state. This is the assumption the present framework relaxes. Instead:

$$\Psi(\mathbf{r},t) = P_{S_3 \cup S_4}(\mathcal{I}) \quad \text{with} \quad \mathcal{I} \supset P_{S_3 \cup S_4}(\mathcal{I}) \tag{4}$$

The residual is:

$$\mathcal{R}_{S_3} = \mathcal{U}_{S_3}(\mathcal{I}) - \mathcal{E}_{S_3}(\Psi) \tag{5}$$

where $\mathcal{U}_{S_3}$ is the full unprojected identity content at scale S_3 and $\mathcal{E}_{S_3}$ is the classical expectation recovery.

§2 Identity Structure and Scale Ladder

2.1 Scale Ladder Definition

Definition 2.1 (Scale entry)**.** Each scale level S_n is a triple:

$$S_n = (D_n,\ M_n,\ L_n) \tag{6}$$

where D_n is the degree of freedom / dimensional permission added at scale n, M_n is the associated mediator or force carrier, and L_n is the observational limit (the scale below which S_n becomes relevant).

Scale	Meaning	Mediator	Observational limit
S_0	Binary existence $E \in \{0, 1\}$	—	Planck?
S_1	Linear vector permission	—	Sub-nuclear
S_2	Relational / angular	Gluons (color)	Nuclear
S_3	Spatial / binding structure	Photons (EM)	Atomic
S_4	Time-ordered transformation	W, Z bosons	Weak
S_5	Mass-energy curvature	Graviton (hypothetical)	Cosmological
S_D	Dark-scale permission w	X_D (unknown)	L_{EM}

Table 1: Scale ladder. Time t is the sequencing parameter of $\mathcal{I}(t)$, not a scale level itself.

2.2 Identity Bundle and Identity Anchor

A particle is not a point. It is an identity bundle:

$$\mathcal{I}_i(t) = \{c_{i1}, c_{i2}, \dots, c_{im}\} \tag{7}$$

where each component c_{ik} carries a scale-level tag.

The identity anchor (generalized center) is:

$$\mathcal{A}_i(t) = \frac{\sum_k \alpha_{ik}\, c_{ik}(t)}{\sum_k \alpha_{ik}} = \frac{\boldsymbol{\alpha}_i^\top \mathbf{C}_i}{\boldsymbol{\alpha}_i^\top \mathbf{1}} \tag{8}$$

The observed 3D position is only the projection:

$$\vec{r}_i^{\,\text{obs}} = P_{3D}(\mathcal{A}_i) \tag{9}$$

The fuzz radius measures internal spread at the observed scale:

$$r_f^2 = \frac{\sum_k \alpha_k \left\| P_{3D}(c_k) - \vec{r}^{\,\text{obs}} \right\|^2}{\sum_k \alpha_k} \tag{10}$$

Definition 2.2 (Fuzz ratio)**.**

$$\Phi_S = \frac{r_f}{\ell_S} \tag{11}$$

where ℓ_S is the characteristic length of scale S.

$$\begin{aligned} \Phi_S \ll 1 &\implies \text{point-like at scale } S \\ \Phi_S \sim 1 &\implies \text{fuzz-zone} \\ \Phi_S > 1 &\implies \text{non-point-defined at scale } S \end{aligned}$$

§3 The Alternative Wavefunction

3.1 Extended State Space

In the standard framework, the state space is:

$$\Psi : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{C} \tag{12}$$

In the extended framework, the full identity state lives in:

$$\Psi_{\text{full}} : \mathbb{R}^3 \times w \times \mathbb{R} \to \mathbb{C} \qquad \Psi_{\text{full}} = \Psi(x, y, z, w;\, t) \tag{13}$$

where w is the dark-scale coordinate. The observed wavefunction is the projection:

$$\Psi_{\text{obs}}(\mathbf{r}, t) = \int \Psi_{\text{full}}(\mathbf{r}, w, t)\, K_{\text{EM}}(w)\, dw \tag{14}$$

with $K_{\text{EM}}(w)$ the electromagnetic projection kernel.

3.2 The Extended Schrödinger Equation

Replace $\hat{H}$ with the extended Hamiltonian:

$$\hat{H}_{\text{eff}} = \hat{H}_{S_3} + \hat{H}_{S_4} + \hat{H}_w \tag{15}$$

where $\hat{H}_w$ is the dark-scale Hamiltonian term. The extended equation becomes:

$$\boxed{i\hbar \frac{\partial}{\partial t}\Psi_{\text{full}}(\mathbf{r}, w, t) = \hat{H}_{\text{eff}}\, \Psi_{\text{full}}(\mathbf{r}, w, t)} \tag{16}$$

Integrating out w with kernel $K_{\text{EM}}(w)$ recovers (1) exactly when $\hat{H}_w$ is negligible, plus a correction term:

$$i\hbar \frac{\partial \Psi_{\text{obs}}}{\partial t} = \hat{H}_{S_3}\, \Psi_{\text{obs}} + \underbrace{\int \hat{H}_w\, \Psi_{\text{full}}\, K_{\text{EM}}(w)\, dw}_{\delta \hat{H}_w[\Psi_{\text{obs}}]} \tag{17}$$

Remark 3.1. $\delta \hat{H}_w$ is the residual Hamiltonian term invisible to EM projection. It encodes exactly the physics that the standard equation misses. In regimes where $\hat{H}_w \approx 0$, standard QM is recovered without modification.

3.3 Cancellation and Hidden Active Structure

A critical property of the extended framework is that observable channel cancellation does not imply internal inactivity.

Define the projected observable I_j and hidden intensity H_j for component j:

$$I_j = \boldsymbol{\alpha}^{\top} \mathbf{c}_j \tag{18}$$

$$H_j = \mathbf{c}_j^{\top} \mathbf{W}\, \mathbf{c}_j \tag{19}$$

The hidden-active condition:

$$\boxed{I_j = 0, \quad H_j > 0} \tag{20}$$

Example: neutron-like charge cancellation.

$$I_{\text{charge}} = +\tfrac{2}{3} - \tfrac{1}{3} - \tfrac{1}{3} = 0 \tag{21}$$

$$H_{\text{charge}} = \left(\tfrac{2}{3}\right)^2 + \left(-\tfrac{1}{3}\right)^2 + \left(-\tfrac{1}{3}\right)^2 = \tfrac{2}{3} \neq 0 \tag{22}$$

The particle appears charge-neutral (zero observable channel) but has nonzero internal charge intensity. This structure is hidden from the K_{EM} projection but present in $\hat{H}_w$.

This generalizes to:

Observable channel	Hidden-active mechanism
Electric charge	Quark charge cancellation
Color charge	Color neutrality (confinement)
Spin	Singlet pairing
Linear momentum	Bound-state internal motion
Angular momentum	$L + S = J = 0$ nuclear ground states
Magnetic moment	Quenched orbital contributions
Defect pairs	Frenkel / Schottky charge balance
Mechanical stress	Residual stress in balanced structures

§4 Projection Failure and Exactness

4.1 Many-to-One Projection

Definition 4.1 (Projection non-injectivity)**.** A projection P_S is non-injective when:

$$P_S(\mathcal{I}_a) = P_S(\mathcal{I}_b) \quad \text{while} \quad \mathcal{I}_a \neq \mathcal{I}_b \tag{23}$$

Theorem 4.1 (Exactness failure)**.** *When P_S is non-injective:*

$$P_S^{-1}(\mathcal{O}_S) \text{ is not unique.} \tag{24}$$

Exact reconstruction from observation at scale S is impossible.

Sketch. Two distinct identity states $\mathcal{I}_a \neq \mathcal{I}_b$ produce identical observations $\mathcal{O}_S$. No measurement at scale S can distinguish them. A unique inverse map would require information not present at scale S. □ □

4.2 Destructive Inverse Packing

Packing is the forward process:

$$\mathbf{C}^{(n)} \xrightarrow{\Pi_n} \mathbf{I}^{(n+1)} \tag{25}$$

The inverse (annihilation / separation):

$$\mathbf{I}^{(n+1)} \xrightarrow{\mathcal{A}} \mathbf{C}^{(n)}_{\text{out}} \tag{26}$$

necessarily *destroys or transforms* the original packed identity.

Proposition 4.1. *Complete hidden-variable maps require destructive inverse packing. Therefore exact mapping cannot preserve the system being mapped.*

This connects to the quantum no-cloning theorem and to the irreversibility of measurement. It is not an epistemological limitation; it is a structural consequence of packing.

§5 Entanglement as a Relation-Particle

5.1 The Relation Object

Two entangled particles A and B share a relation object:

$$\mathcal{R}_{AB} = [0, 0, 0, 0 \mid \mathbf{T}_{AB}] \tag{27}$$

The first four entries are spacetime displacement channels:

$$\Delta x = \Delta y = \Delta z = \Delta t = 0 \tag{28}$$

The trinary relation vector:

$$\mathbf{T}_{AB} \in \{-1, 0, +1\}^m \tag{29}$$

carries the real correlational content.

Remark 5.1. $[0, 0, 0, 0]$ means *no spacetime displacement*, not *no physical existence.* The relation object is real, deterministic, and non-spatiotemporal.

5.2 Pair Decomposition and Measurement

The joint identity decomposes as:

$$\mathcal{I}_{AB} \;\to\; \mathcal{I}_A + \mathcal{I}_B + \mathcal{R}_{AB} \tag{30}$$

Measurement outcomes are scale-projected with the relation object:

$$O_A = P_a(\mathcal{I}_A,\, \mathcal{R}_{AB}) \tag{31}$$

$$O_B = P_b(\mathcal{I}_B,\, \mathcal{R}_{AB}) \tag{32}$$

The correlation function:

$$E(a,b) = \langle O_A\, O_B \rangle \tag{33}$$

reproduces quantum-mechanical predictions because $\mathcal{R}_{AB}$ is a fixed structure in identity-space, not a propagating signal.

§6 Dark Matter as Higher-Scale Projection Residual

6.1 The Projection Gap

$$\rho_{\text{dark}}(x,y,z;\,t) = P_G(\rho) - P_{\text{EM}}(\rho) \tag{34}$$

In kernel form, integrating over the dark coordinate w:

$$\boxed{\rho_{\text{dark}}(x,y,z;\,t) = \int \rho(x,y,z,w;\,t)\,[K_G(w) - K_{\text{EM}}(w)]\,dw} \tag{35}$$

$K_G(w)$ is the gravitational projection kernel (couples to all scales); $K_{\text{EM}}(w)$ is the electromagnetic projection kernel (truncates at L_{EM}).

6.2 Dark-Scale Entry

The dark-scale level in the ladder:

$$S_D = (w,\; X_D,\; L_{\text{EM}}) \tag{36}$$

where w is the dark-scale degree of freedom, X_D is the (presently unknown) dark mediator, and L_{EM} marks where electromagnetic projection fails.

Central claim:

$$\boxed{\text{dark matter} = \text{gravitational signature of an unresolved higher-scale permission}} \tag{37}$$

This is distinct from standard extra-dimensional models:

$$\begin{aligned} \text{Standard:}&\quad \text{dimension first} \to \text{dark behavior} \\ \text{Inverse KK:}&\quad \text{force-scale residuals first} \to \text{inferred dimension} \end{aligned}$$

§7 Effective Proper Time with Dark-Scale Correction

7.1 Classical and Quantum Proper Time

Classical relativistic proper time:

$$d\tau_{\text{4D}} = dt\sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2}} \tag{38}$$

Recent optical ion clock work establishes:

$$\hat{\tau} \equiv \tau(\hat{x}, \hat{p}) \tag{39}$$

Proper time becomes quantum-linked; $\hat{\tau}$ is an observable with spread $R_\tau = \hat{\tau} - \langle \hat{\tau} \rangle$.

7.2 Dark-Scale Extension

Extend to:

$$\boxed{d\tau_{\text{eff}} = dt\sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2} + \chi_w(w)}} \tag{40}$$

where $\chi_w(w)$ is the dark-scale metric correction term.

In differential form:

$$d\tau_{\text{eff}} = d\tau_{\text{4D}} + d\tau_w \tag{41}$$

The full proper-time identity structure:

$$\mathcal{I}_\tau = (\hat{x},\, \hat{p},\, \hat{\tau},\, H_c,\, H_m,\, w) \tag{42}$$

where H_c is the clock Hamiltonian and H_m is the motional Hamiltonian. The clock-motion relation object:

$$\mathcal{R}_{cm} = [0, 0, 0, 0 \mid \mathbf{T}_{cm}] \tag{43}$$

Remark 7.1. $\chi_w(w)$ remains formally undefined here. Constraining it requires either: (a) a specific model for $\hat{H}_w$, or (b) comparison with clock-discrepancy data in strong gravitational fields or extreme velocities where 4D relativity is insufficient.

§8 Recovery of Standard Quantum Mechanics

The critical consistency check: the standard Schrödinger equation must be exactly recovered in the appropriate limit.

Proposition 8.1 (Standard QM recovery). *When $\hat{H}_w \approx 0$ (dark-scale Hamiltonian negligible) and $\chi_w(w) \approx 0$ (proper-time correction negligible), the extended framework reduces exactly to:*

$$\Psi_{obs}(\mathbf{r}, t) \approx \Psi(\mathbf{r}, t) \tag{44}$$

$$i\hbar\, \partial_t \Psi = \hat{H}_{S_3} \Psi \tag{45}$$

which is (1)*.*

The framework is therefore a *conservative extension.* No existing prediction is contradicted; corrections appear only where the extended terms are non-negligible: strong gravity, cosmological scales, and dark-sector phenomenology.

§9 Open Problems

1. **Form of $K_{\mathbf{EM}}(w)$ and $K_G(w)$.** These projection kernels determine the magnitude of all dark-sector corrections. They are presently unconstrained.
2. **Form of $\hat{H}_w$.** Without a specific dark-scale Hamiltonian the theory has no new quantitative predictions.
3. **Form of $\chi_w(w)$.** The dark-scale metric correction in (40) needs a physical derivation or at minimum a phenomenological ansatz.
4. **Trinary relation vector $\mathbf{T}_{AB}$.** The space $\{-1, 0, +1\}^m$ must be given a dynamics. How does $\mathbf{T}_{AB}$ evolve under local operations?
5. **Packing operator Π_n.** The map from $\mathbf{C}^{(n)}$ to $\mathbf{I}^{(n+1)}$ is stated as an axiom. Its explicit form at each scale level is needed.
6. **Connection to existing field theory.** The extended equation (16) must be Lorentz-covariant. The dark coordinate w must respect causality or explain why it does not.

§A Notation Summary

Symbol	Meaning
$\mathcal{I}$	Full identity state
P_S	Projection onto scale S
$\mathcal{R}_S$	Residual at scale S: $\mathcal{U}_S(\mathcal{I}) - \mathcal{E}_S(\mathcal{O}_S)$
$S_n = (D_n, M_n, L_n)$	Scale entry: DOF, mediator, observational limit
w	Dark-scale degree of freedom
$\mathcal{A}_i$	Identity anchor (eq. 8)
r_f	Fuzz radius (eq. 10)
Φ_S	Fuzz ratio
Π_n	Packing operator
$\mathcal{R}_{AB}$	Relation-particle object (eq. 27)
$\mathbf{T}_{AB}$	Trinary relation vector
$K_G(w), K_{\mathrm{EM}}(w)$	Gravitational / EM projection kernels
$\chi_w(w)$	Dark-scale metric correction
$d\tau_{\mathrm{eff}}$	Effective proper time (eq. 40)